

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

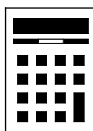
Paper 1

Friday 3 November 2023 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



Answer **all** questions in the spaces provided.

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outside the
box

0 1

This question is about the element aluminium.

An aluminium atom can be represented as $^{27}_{13}\text{Al}$

0 1 . 1

What is the mass number of this aluminium atom?

[1 mark]

Tick (✓) **one** box.

13

☐

14

☐

27

☐

40

☐

0 1 . 2

How many protons are there in an aluminium atom?

[1 mark]

Tick (✓) **one** box.

13

☐

14

☐

27

☐

40

☐

0 1 . 3 What is the relative charge of a proton?

[1 mark]

Tick (✓) **one** box.

+1

☐

0

☐

−1

☐

0 1 . 4 Name an element that has similar **chemical** properties to aluminium.

Give **one** reason for your choice.

Use the periodic table.

[2 marks]

Element _____

Reason _____

Question 1 continues on the next page

Turn over ►



Aluminium reacts with chlorine to produce aluminium chloride.

0 1 . 5

Write a **word** equation for the reaction of aluminium with chlorine.

[1 mark]

_____ + _____ → _____

0 1 . 6

Aluminium reacts with 213 g of chlorine to produce 267 g of aluminium chloride.

Determine the mass of aluminium that reacts with 213 g of chlorine.

[1 mark]

Mass of aluminium = _____ g

0 1 . 7

A compound containing aluminium has the formula AlXO_4

X is not the symbol of the element.

The relative formula mass of the compound is 122

Determine the identity of element **X**.

Relative atomic masses (A_r): Al = 27 O = 16

[3 marks]

Element **X** = _____

10



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

Turn over ►



0 2

Crude oil is a mixture of many different compounds.

Crude oil contains alkanes.

Alkanes are made up of hydrogen and carbon atoms only.

0 2 . 1

What is the name of compounds made up of hydrogen **and** carbon atoms only?

[1 mark]

Tick (✓) **one** box.

Carbonates

☐

Hydrocarbons

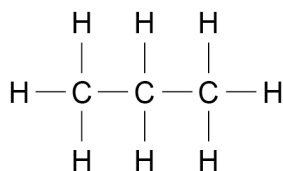
☐

Hydroxides

☐**0 2 . 2**

Figure 1 shows an alkane.

Figure 1



What is the name of the alkane in **Figure 1**?

[1 mark]

Tick (✓) **one** box.

Ethane

☐

Methane

☐

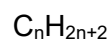
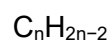
Propane

☐

0 2 . 3 Which is the general formula of alkanes?

[1 mark]

Tick (✓) **one** box.

☐☐☐☐

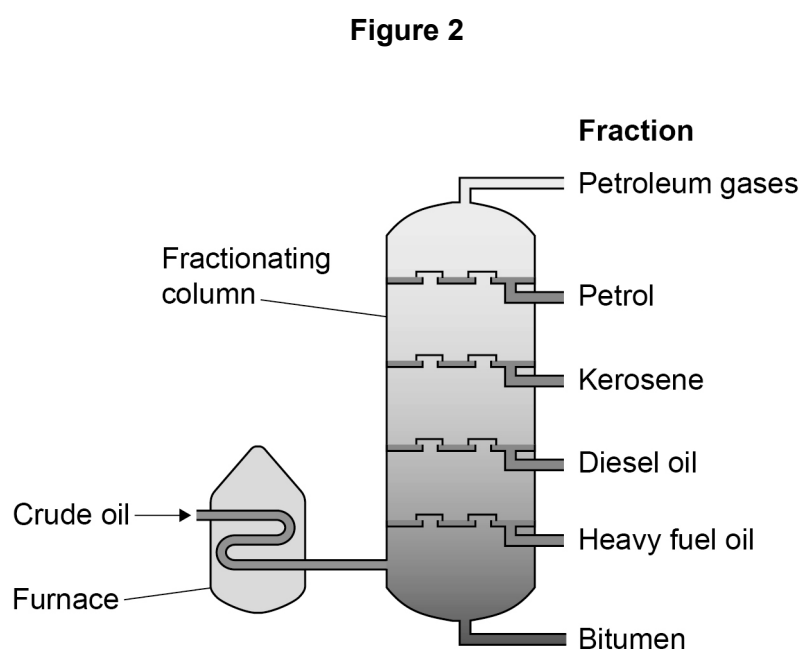
Question 2 continues on the next page

Turn over ►



Crude oil is separated into fractions by fractional distillation.

Figure 2 shows a fractionating column used to separate crude oil.



0 2 . 4

Describe how crude oil is separated into fractions by fractional distillation.

[4 marks]



0	2	.	5
---	---	---	---

Most of the fractions in **Figure 2** are used as fuels.

Bitumen is **not** used as a fuel.

Suggest **two** reasons why.

[2 marks]

1 _____

2 _____

Question 2 continues on the next page

Turn over ►



Table 1 shows the percentages (%) of different fractions by mass in a sample of crude oil.

Table 1

Fraction	Percentage of fraction by mass
Petroleum gases	14.7
Petrol	28.6
Kerosene	20.5
Diesel oil	X
Heavy fuel oil	12.0
Bitumen	8.8

0 2 . 6 Determine the percentage of diesel oil (**X**) in the sample of crude oil.

Use **Table 1**.

[1 mark]

Percentage of diesel oil = _____ %



0 2 . 7 The sample of crude oil has a mass of 143 kg.

Determine the mass of kerosene in this sample.

Give your answer to 3 significant figures.

Use **Table 1**.

[3 marks]

Mass of kerosene (3 significant figures) = _____ kg

The petrol fraction contains octane (C_8H_{18}).

The equation for the incomplete combustion of octane is:



0 2 . 8 Calculate the total number of moles of gas produced from the incomplete combustion of 0.75 moles of octane.

[2 marks]

Moles of gas produced = _____

0 2 . 9 Give **one** reason why octane may react by incomplete combustion rather than complete combustion.

[1 mark]



0 3

This question is about identifying ions.

Flame tests can be used to identify metal ions.

A student tested lithium carbonate using a flame test.

This is the method used.

- 1 Place a metal wire in hydrochloric acid.
- 2 Place the metal wire in lithium carbonate powder.
- 3 Put the metal wire in the blue flame of a Bunsen burner.
- 4 Record the flame colour produced.

0 3 . 1

Why was the metal wire placed in hydrochloric acid?

[1 mark]

0 3 . 2

Which flame colour is produced by lithium carbonate?

[1 mark]

Tick (✓) **one** box.

Crimson

☐

Green

☐

Lilac

☐

Yellow

☐

0	3	.	3
---	---	---	---

Lithium carbonate decomposes on heating to produce a white solid and carbon dioxide gas.

Why do lithium carbonate **and** the white solid both produce the **same** colour in a flame test?

[1 mark]

0	3	.	4
---	---	---	---

Describe a test to show that lithium carbonate contains carbonate ions.

Give the result of the test.

[2 marks]

Test

Result

Question 3 continues on the next page

Turn over ►



0 3 . 5 Lithium carbonate has the formula Li_2CO_3

Calculate the number of moles of **lithium ions** in 3.70 g of lithium carbonate.

Relative formula mass (M_r): $\text{Li}_2\text{CO}_3 = 74$

[3 marks]

Moles of lithium ions = _____

0 3 . 6 The student dissolved a different lithium compound in water to produce a solution.

The student added dilute nitric acid and silver nitrate solution to the solution of the lithium compound.

A yellow precipitate was formed.

Identify the lithium compound.

Give **one** reason for your answer.

[2 marks]

Lithium compound _____

Reason _____

10



0	4
---	---

This question is about Group 1 elements.

0	4	.	1
---	---	---	---

What name is given to the elements in Group 1 of the periodic table?

[1 mark]

Tick (✓) **one** box.

Alkali metals

☐

Halogens

☐

Noble gases

☐

Transition metals

☐

0	4	.	2
---	---	---	---

Which is a property of Group 1 elements?

[1 mark]

Tick (✓) **one** box.

Are soft and easily cut

☐

Form coloured compounds

☐

Have ions with different charges

☐

Useful as catalysts

☐

Question 4 continues on the next page

Turn over ►



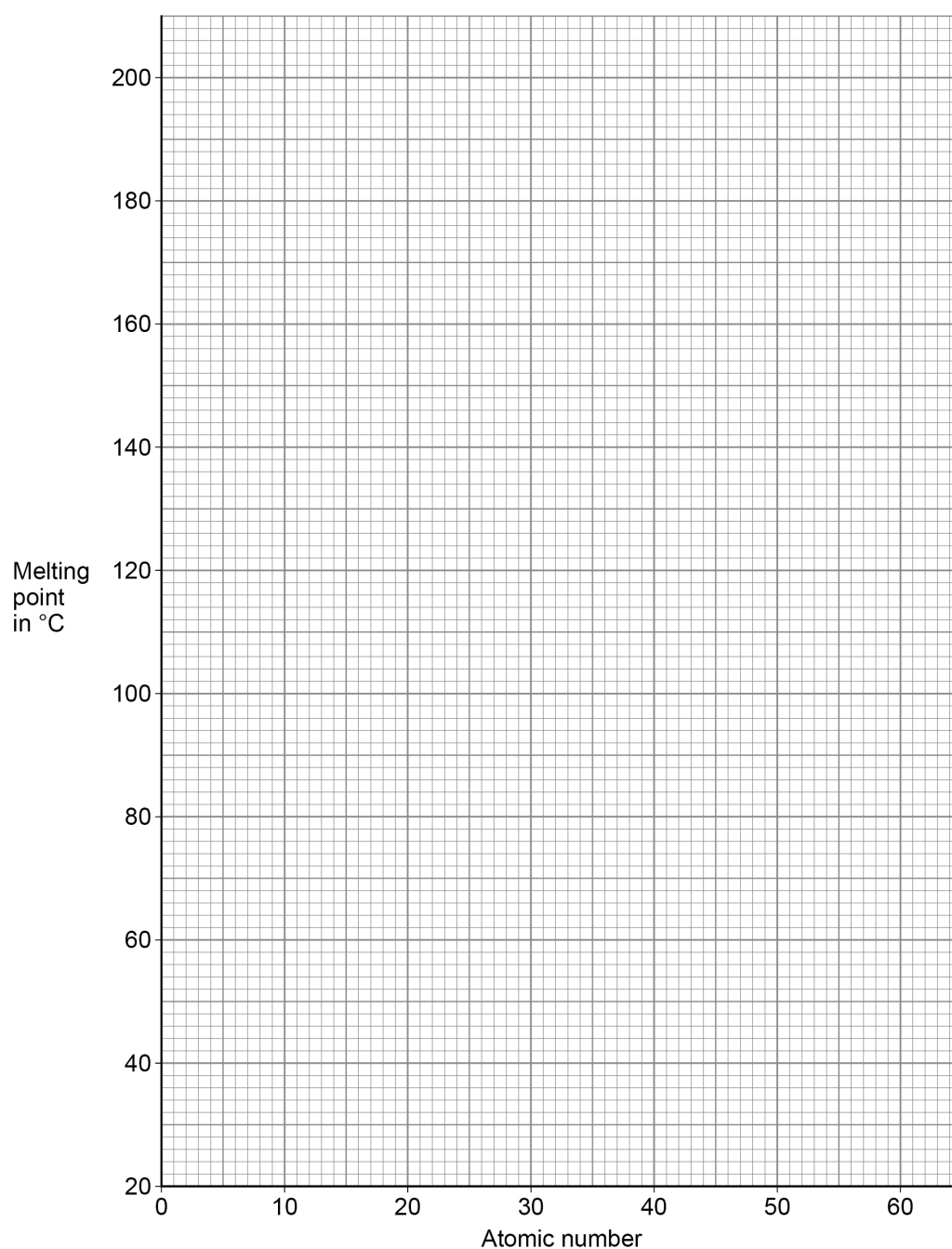
Table 2 shows the atomic numbers and the melting points of some Group 1 elements.

Table 2

Element	Atomic number	Melting point in °C
Lithium	3	181
Sodium	11	98
Rubidium	37	39
Caesium	55	29

Figure 3 shows a graph of atomic number and melting point in °C.

Figure 3



0 4 . 3 Complete **Figure 3**.

You should plot the atomic number and melting point data from **Table 2** on **Figure 3**.
[2 marks]

0 4 . 4 Potassium has an atomic number of 19

Predict the melting point of potassium.

Use **Table 2** and **Figure 3**.

[1 mark]

Melting point = _____ °C

Question 4 continues on the next page

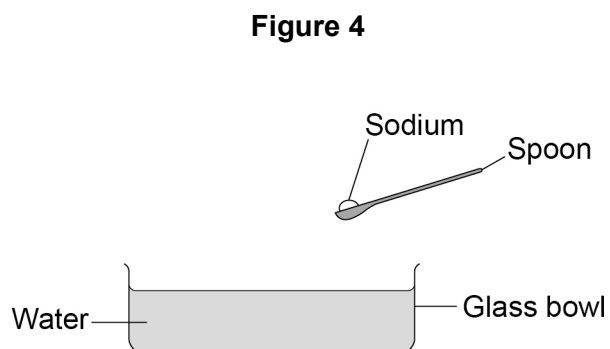
Turn over ►



Group 1 elements react with water.

A teacher added a small piece of sodium to water.

Figure 4 shows the apparatus.



The sodium floated on the surface of the water.

0 4 . 5

Give **one** other observation made when a small piece of sodium is added to water.

[1 mark]

0 4 . 6

A small piece of rubidium does **not** float on the surface of the water.

Why does rubidium **not** float on water?

[1 mark]

0 4 . 7

Potassium reacts with water to produce potassium hydroxide and hydrogen.

Complete the equation for the reaction of potassium with water.

You should balance the equation.

[3 marks]



0 4 . 8

A solution of potassium hydroxide contained 0.300 moles of potassium hydroxide dissolved in 0.600 dm³ of water.

Calculate the mass of potassium hydroxide in 45.0 cm³ of this solution.

The relative formula mass (M_r) of potassium hydroxide is 56

[4 marks]

Mass of potassium hydroxide = _____ g

0 4 . 9

Explain how the reactivity of the elements changes going down Group 1.

[4 marks]

18

Turn over for the next question

Turn over ►

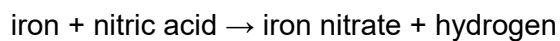


0 5

This question is about acids and salts.

Iron nitrate solution is produced when iron reacts with nitric acid.

The word equation for the reaction is:

**0 5 . 1**

What is the pH of nitric acid?

[1 mark]Tick (✓) **one** box.

1

☐

7

☐

10

☐

14

☐**0 5 . 2**

What is the formula of nitric acid?

[1 mark]Tick (✓) **one** box.

HCl

☐HNO₃☐H₂SO₄☐

0 5 . 3 What is the state symbol for iron nitrate solution?

[1 mark]

Tick (✓) **one** box.

(aq)

☐

(g)

☐

(l)

☐

(s)

☐

Question 5 continues on the next page

Turn over ►



0	5	.	4
---	---	---	---

The formula of a sulfate ion is SO_4^{2-}

[1 mark]

0	5	.	5
---	---	---	---

[6 marks]

[illegible]

10



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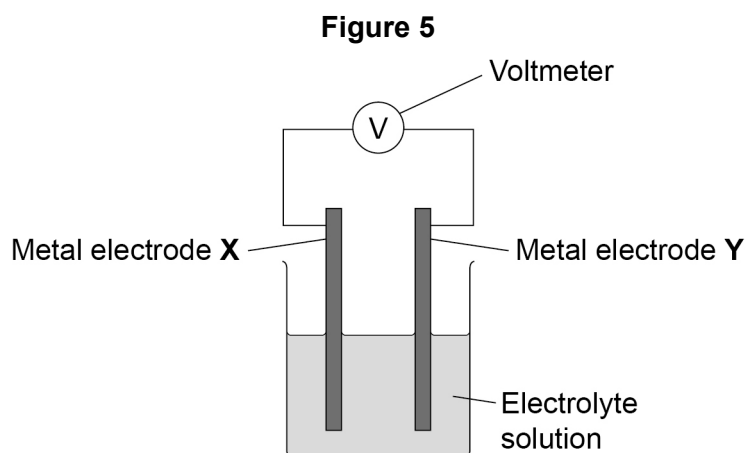
Turn over ►



0 6

A student investigated the potential difference produced by chemical cells. The student changed one of the metal electrodes.

Figure 5 shows the apparatus.



The student:

- used a voltmeter to measure the potential difference of the chemical cell in volts
- used the **same** metal each time for electrode **X**
- repeated the experiment using a **different** metal each time for electrode **Y**.

0 6 . 1

Give **one** control variable the student should keep the same to make the investigation valid.

Do **not** refer to the electrodes.

[1 mark]

Table 3 shows the results.

Table 3

Electrode Y	Potential difference of the chemical cell in volts
Cobalt	+0.09
Copper	-0.59
Magnesium	+2.12
Nickel	0.00
Vanadium	+0.90



0 6 . 2 Which metal is electrode **X**?

Give **one** reason for your answer.

[2 marks]

Metal _____

Reason _____

0 6 . 3 Write the five metals used for electrode **Y** in order of reactivity.

Use **Table 3**.

Give **one** reason for your answer.

[3 marks]

Most reactive _____

Least reactive _____

Reason _____

0 6 . 4 Which **two** metals in **Table 3** would produce a potential difference of 1.49 V when used as the electrodes in a chemical cell?

Use a calculation to justify your answer.

[3 marks]

Electrode **X** _____

Electrode **Y** _____

Justification _____

Question 6 continues on the next page

Turn over ►



0 6 . 5

The student repeated the experiment using aluminium as electrode Y.

Aluminium atoms lose electrons to produce aluminium ions.

Write a half equation for the reaction at electrode Y.

[2 marks]



0 6 . 6

Give **one** reason why the potential difference produced by a chemical cell eventually becomes zero.

[1 mark]

12

0 7

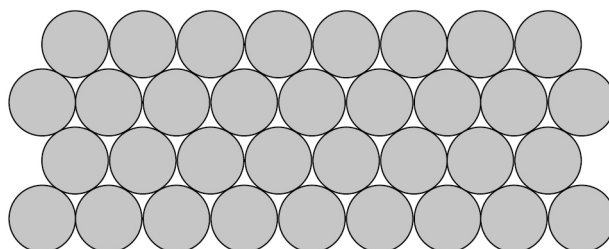
This question is about magnesium.

Magnesium has a giant structure held together by metallic bonding.

0 7 . 1

Explain how magnesium conducts heat.

[3 marks]

0 7 . 2**Figure 6** represents the structure of magnesium.**Figure 6**

Explain why magnesium can be bent or hammered into shape.

[2 marks]

Question 7 continues on the next page**Turn over ►**

Magnesium reacts with fluorine to produce the ionic compound magnesium fluoride.

0 7 . 3

Describe what happens when magnesium atoms react with fluorine atoms to produce magnesium fluoride.

[4 marks]

0 7 . 4

Explain why magnesium fluoride has a high melting point.

You should answer in terms of structure and bonding.

[4 marks]



0 7 . 5 **Table 4** shows the melting points of two magnesium compounds.

Table 4

	Melting point in °C
Magnesium fluoride	1263
Magnesium oxide	2852

Suggest **one** reason why the melting point of magnesium oxide is higher than the melting point of magnesium fluoride.

[1 mark]

14

END OF QUESTIONS



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Question number	Additional page, if required. Write the question numbers in the left-hand margin.
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